

Autodesk® Scaleform®

CRI

Scaleform Video

CRI

Vladislav Merker
1.02
2013 1 18

Copyright Notice

Autodesk® Scaleform® 4.3

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER

EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

	CRI
	Scaleform Corporation
	6305 Ivy Lane, Suite 310
	Greenbelt, MD 20770, USA
	www.scaleform.com
Email	info@scaleform.com
	+1 (301) 446-3200
Fax	+1 (301) 446-3199

Table of Contents

1 1

2 / 1

2.1 1

2.1.1 1

2.1.2 2

2.2 3

2.3 3

2.3.1 3

2.3.2 4

2.4 4

2.4.1 4

2.4.2 5

3 CRI 6

3.1 6

3.1.1 6

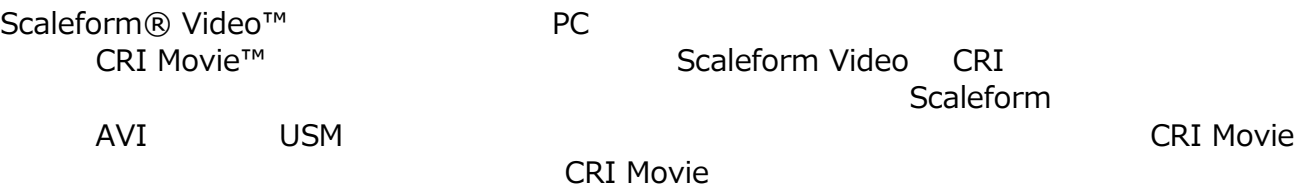
3.1.2 6

3.1.3 Video 7

3.1.4 7

3.1.5		7
3.2		7
3.2.1		7
3.2.2		11
3.2.3	Video	11
3.2.4	GOP Group of Pictures	13
3.2.5		14
3.2.6		15

1 はじめに



2 入力/出力ファイル

- 入力ファイル
 -
 -
 -
 -
- 出力ファイル
 - CRI .usm

2.1 ビデオ素材ファイル

2.1.1ビデオ素材ファイルの仕様

AVI

推奨ビデオ素材の仕様	
項目	説明
	Microsoft AVI 2GB
	PCM WAVE
	RGB BMP

ビデオ素材の仕様	
項目	説明
	AVI (*1).
	24 /32
	[] 72×72 [] 4088×4088 8×8 (*2)
	60 /59.94 /50 /30 /29.97 /25 /24 /23.976 fps (*3).
	[] (*4) [] D1

[]

(*1) DirectShow Windows Media Player
AVI AVI

AVI

DirectShow

(*2) PlayStation 2 16×16

(*3) 29.97fps
15fps

(*4) Wii

2.1.2連番付き静止画イメージファイル仕様

CRI

ビデオ素材の仕様	
項目	説明
	1) 24 /32 BMP 2) 24 /32 TGA /
	[] 72×72 [] 8×8
	1) " " " " " " OK> abc001.bmp, abc002.bmp, abc003.bmp , ... NG> abc001def.bmp, abc002def.bmp, abc003def.bmp, ... 2) OK> abc998.bmp, abc999.bmp, abc1000.bmp, ... 3)

	OK> abc001.bmp, abc004.bmp, abc008.bmp, ...
--	---

2.2 オーディオ素材ファイル

- AVI
-

推奨オーディオ素材ファイルの仕様	
項目	説明
	WAVE AIFF
	16 PCM
	48kHz

2.3 字幕情報ファイル

2.3.1 字幕情報ファイルのフォーマット

- 1
- 2 , INTIME
- OUTTIME
-
-
- ;
- ASCII Shift JIS UTF-8 UTF-16 UTF-16LE

2.3.2 字幕情報ファイルの例

例 1

1000 0 2 5 7.5 7.5 10

```
;INTIME, OUTTIME, Subtitle, Comment
1000
0, 2000, FirstMessage.
5000, 7500, Second Message.
7500, 10000, Third Message.
```

例 2

29.97fps 29970 1 1000
 $29.97 \times 1000 = 29970$
 29970 0 100 150 200
 200 250

```
;INTIME, OUTTIME, Subtitle, Comment
29970
0, 100000, FirstMessage.
150000, 200000, Second Message.
200000, 250000, Third Message.
```

2.4 キューポイント情報ファイル

2.4.1 キューポイント情報ファイルのフォーマット

1.	1				
2.	2	,		Time	Value
		EventPointName	String		
3.		Time			Time
4.	2		Value		
5.	3		EventPointName		

6. 4 String

7. ;

8. ASCII Shift JIS UTF-8 UTF-16 UTF-16LE

2.4.2キューポイント情報ファイルの例

例 1

```
          1000
0  5  7  8 15 25
          #3  #4
```

```
;Time, Value, EventPointName, String
1000
0,0,navi1
5000,0,navi2
7000,1,evpt1,param,test
8000,1,evpt2,param1,300,param2,500
15000,0,navi3
15000,1,evpt3
25000,0,navi4
```

3 CRI エンコーダの基本的な使い方

MS-DOS

```
[      ] medianoche [-subcommand=param1[,param2,...]]
[      ] subcommand :
          param#    :
```

AVI sample.avi CRI sample.usm

```
> medianoche -in=sample.avi -out=sample.usm
```

3.1 サブコマンド

3.1.1 ファイルサブコマンド

サブコマンド	説明
in	
out	
video##	
alpha##	
audio##	
mca**_##	
subtitle##	
cuepoint	

3.1.2 ビデオ編集サブコマンド

サブコマンド	説明
crop	
scale	/


```
[ ] -in=filename
[ ] filename :
[ ]
```

sample.avi

```
> medianoche -in=sample.avi -out=sample.usm
```

out: 出カムービーファイルを指定します

```
[ ] -out=filename
[ ] output : usm
[ ]
```

.usm

sample.usm

```
> medianoche -in=sample.avi -out=sample.usm
```

video: 入力ビデオファイルを指定します

```
[ ] -video##=filename
[ ] filename :
[ ] ## : (## = 00)
```

1. セクション 2.1
2. 00

AVI *smpvid.avi* 00

```
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm
```

alpha: アルファファイルを指定します

```
[ ] -alpha##=filename
[ ] filename :
[ ] ## : (## = 00)
```

AVI

1. 00

```
AVI      smpvid.avi      00
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm \
  -alpha00=smpvid.avi
```

audio: オーディオファイルを指定します

```
[      ] -audio##=filename
[      ] filename :
          ##      : (## = 00 - 31)
[      ]
```

1. セクション 2.2

2. • CRI

• 00

• 32

00 01 02

```
> medianoche -video00=smpvid.avi -out=sample.usm \
  -audio00=music.wav -audio01=voice_en.wav -audio02=voice_ja.wav
```

mca: サラウンドムービー用のオーディオファイルを指定します

```
[      ] -mca**_##=filename
[      ] filename :
          **      :
          ##      : (## = 00 - 05)
[      ]
```

1. •

•

2. •

•

チャンネル番号	スピーカーポジション
00	
01	
02	
03	
04	
05	LFE

0

```
> medianoche -video00=smpvid.avi -out=sample.usm \
    -mca00_00=fl.wav -mca00_01=fr.wav -mca00_02=sl.wav -mca00_03=sr.wav \
    -mca00_04=c.wav -mca00_05=lfe.wav
```

subtitle: 字幕情報ファイルを指定します

```
[          ] -subtitle##=filename
[  ]      filename :
          ##      :          (## = 00 - 15)
```

1. セクション 2.3
2. 16

00 01

```
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm \
    -subtitle00=subttl_en.txt -subtitle01=subttl_ja.txt
```

cuepoint: キューポイント情報ファイルを指定します

```
[          ] -cuepoint=filename
[  ]      filename :
[  ]
```

セクション 2.4

```
> medianoche -in=sample.avi -out=sample.usm -cuepoint=sample_cuepoint.txt
```

3.2.2ビデオ編集サブコマンド

crop: オリジナルイメージの一部をクロップします

```
[ ] crop=crx,cry,cw,crh
[ ] crx : X
    cry : Y
    cw :
    crh :
[ ]
```

0,0 640 480

```
> medianoche -in=sample.avi -out=sample.usm -crop=0,0,640,480
```

scale: オリジナルイメージを拡大/縮小します

```
[ ] scale=scw,sch
[ ] scw :
    sch :
[ ]
```

640 480

```
> medianoche -in=sample.avi -out=sample.usm -scale=640,480
```

3.2.3 Video エンコード設定サブコマンド

bitrate: ビットレート

```
[ ] -bitrate=rate
[ ] rate : [bps]
[ ] bps
```

3000000bps

6Mbps

```
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm \
-bitrate=6000000
```

br_range: 可変ビットレートコントロールでのビットレート範囲

```
[ ] -br_range=min,max
[ ] min : [bps]
    max : [bps]
[ ]
```

min=0bps max=3000000bps

min=0bps max=3000000bps

```
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm \
- br_range=0,3000000
```

framerate: フレームレート

```
[ ] -framerate=decfr
    -framerate=fr_num,fr_den
[ ] decfr : 10
    fr_num :
    fr_den :
[ ] (fps)
```

```
2
1. 10 -framerate=29.97
2. -framerate=30000/1001
```

60fps

29.97fps

```
> medianoche -video00=smpvid.avi -audio00=smpaud.wav -out=sample.usm \
-framerate=29.97
```

hcfilter: 高圧縮フィルタ

```
[ ] -hcfilter=fltype
[ ] fltype :
```

ON :
 OFF :
 AUTO :

[]

3.2.4 GOP (Group of Pictures) サブコマンド

GOP

GOP

GOP

GOP

- I I
- P P
- B B
P

GOP

I P
 GOP I

gop_i: GOP (group of picture) 内の I ピクチャ数

[] -gop_i=count
 [] count : GOP I
 [] GOP I

GOP GOP I P B

- I I
- 1

gop_p: GOP (group of picture) 内の P ピクチャ数

[] -gop_p=count
 [] count : GOP P
 [] GOP P

GOP GOP I P B

- P I B

- 4

gop_b: GOP (group of picture) 内の B ピクチャ数

```
[ ] -gop_b=count
[ ] count : GOP B
[ ] GOP B
```

GOP GOP I P B

- B

- 2

gop_closed: クローズド GOP

```
[ ] -gop_closed=sw
[ ] sw : GOP
      ON : GOP ( )
      OFF : GOP
[ ] GOP
```

GOP GOP GOP

detectsc: シーン変化検出

```
[ ] -detectsc=sw
[ ] sw :
      ON : ( )
      OFF :
```

I I

3.2.5 オーディオエンコード設定サブコマンド

hca: HCA コーデック用にオーディオソースをエンコードします

```
[Format] -hca=sw
[Input] sw: switch for HCA encoding
          ON: Encode as HCA
          OFF: Not encode as HCA (Default)
[Function] Specifies whether or not to encode audio source as HCA codec.
```

hca_quality: HCA としてエンコードする場合のサウンド品質。

[Format] -hca_quality=count
[Input] count: Selects encoding quality from the following 5 values
 5 = Highest
 4 = High (default)
 3 = Medium
 2 = Low (high compression)
 1 = Lowest (highest compression)
[Function] Set sound quality for HCA encoding.

品質	モノ	ステレオまたはそれ 以上
5	1/4	1/4
4	1/6	1/6
3	1/8	1/8
2	1/10	1/12
1	1/12	1/16

3.2.6その他のサブコマンド

-ms_fullpel_still: 静止部分に対するフル pel での動き検索

[] -ms_fullpel_still=sw
[] sw : pel
 ON :
 OFF : ()
[] pel

 pel

cleanup: 中間ファイルの削除

```
[ ] -cleanup=sw
[ ] sw :
    ON : ( )
    OFF :

[ ]

OFF CRI .usm CRI
.usv CRI .usa
ON
```

work_dir: ワーキングディレクトリの設定

```
[ ] -work_dir=dirname
[ ] dirname :
[ ]
```

debug: デバッグ情報の出力

```
[ ] -debug=sw
[ ] sw :
    ON :
    OFF : ( )

[ ]

OFF
```

preview: プレビュー画面の表示

```
[ ] -preview=sw
[ ] sw :
    ON : ( )
    OFF :

[ ]

ON
```